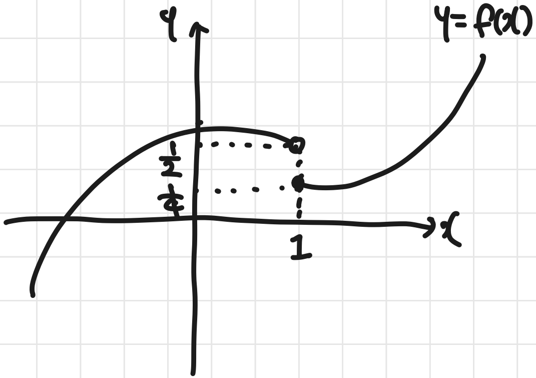
# 함수의 극한

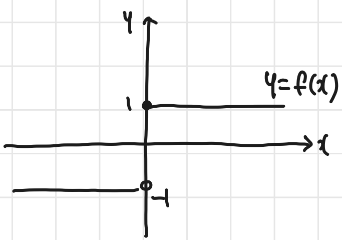
- 좌극한 & 우극한




- 좌극한 $\lim_{x \to 1-} f(x) = \frac{1}{2}$

- 우극한 $\lim_{x \to 1+} f(x) = \frac{1}{4}$

# 예제-20)




$\lim_{x \to 0+} f(x) = 1.$

$\lim_{x \to 0-} f(x) = -1$

- 극한값이 존재할 때

: 함수 $f(x)$가 $\lim_{x\to a+} f(x) = \lim_{x\to a-} f(x) = L$ 일 때,

$L$을 극한(값)이라고 한다.

⇒ $\lim_{x\to a} f(x) = L$ or $x\to a$ 일 때, $f(x)\to L$

- 함수의 극한을 계산하는 방법

1) 대입

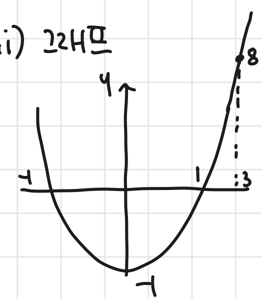
2) 그래프 그려서 (가능하면)

#예13-21)

(a) $\lim_{x\to 3} (x^2-1)$ $(x-1)(x+1)$

i) 대입. : $9-1=8$.

ii) 그래프




$y=(x+1)(x-1)$

$0=(x+1)(x-1)$

$x=-1$ or $1$

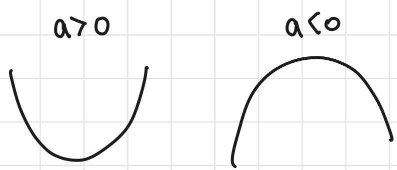
$(-1,0)$ $(1,0)$

(b) $\lim_{x\to -3} \frac{x^2-9}{x+3}$

$= \lim_{x\to -3} \frac{(x+3)(x-3)}{x+3}$

$= -6$

☆ 이차함수 $y=ax^2$



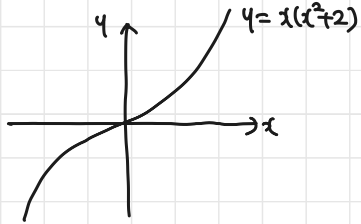
# 예제 2-22)

(a) $\lim_{x \to \infty} \frac{1}{x} = 0$

(b) $\lim_{x \to -\infty} (x^3 + 2x) = \lim_{x \to -\infty} x(x^2+2)$. $(-\infty)^3 + 2 \cdot (-\infty) = -\infty$.

$y = x(x^2+2)$

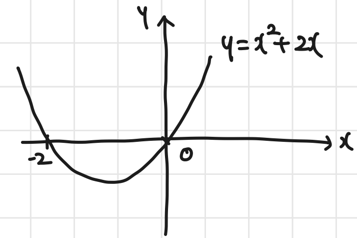
$0 = x(x^2+2)$ $\quad x = 0$




추가 $\lim_{x \to -\infty} (x^2 + 2x) = (-\infty)^2 + 2 \cdot (-\infty)$

$= \infty$

$y = x^2 + 2x = x(x+2)$




- 함수의 극한의 성질

(1) $\lim_{x \to a} c f(x) = c \underbrace{\lim_{x \to a} f(x)}_{} = c\alpha$ $\quad$ ($\lim_{x \to a} f(x) = \alpha$)

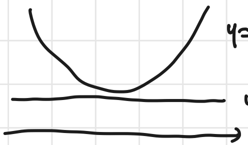
(2) $\lim_{x \to a} \{ f(x) \pm g(x) \} = \underbrace{\lim_{x \to a} f(x)}_{\alpha} \pm \underbrace{\lim_{x \to a} g(x)}_{\beta} = \alpha \pm \beta$

(3) $\lim_{x \to a} \{ f(x) \cdot / g(x) \} = \lim_{x \to a} f(x) \cdot / \lim_{x \to a} g(x) = \alpha\beta \; , \; \alpha/\beta$

• 함수의 극한의 대소관계

: $\lim_{x\to a} f(x)=\alpha$ , $\lim_{x\to a} g(x)=\beta$ 일 때

i) $f(x)\le g(x)$ 이면, $\alpha\le\beta$




ii) $h(x)$가 $f(x)\le h(x)\le g(x)$ 이고, $\alpha=\beta$ 이면, $\lim_{x\to a} h(x)=\overset{\beta}{\alpha}$ 이다.

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# 예13-23).

$$\lim_{x\to3}\frac{\{g(x)\}^3-3f(x)}{\{f(x)\}^2}=\frac{2^3-3\cdot5}{5^2}=\frac{8-15}{25}=\frac{-7}{25}=-\frac{7}{25}$$

# 예13-24)

$$\frac{3x^2-2}{x^2+2}\le\frac{(x^2+2)f(x)}{x^2+2}\le\frac{3x^2+2}{x^2+2}$$

$$\overset{3}{\nwarrow}\lim_{x\to\infty}\frac{3x^2-2}{x^2+2}\le\lim_{x\to\infty}f(x)\le\lim_{x\to\infty}\frac{3x^2+2}{x^2+2}\nearrow 3$$

$\therefore \lim_{x\to\infty} f(x)=3$

#예13-25)

(a) $$\lim_{x \to -1} \frac{x^2 - x - 2}{x+1} = \lim_{x \to -1} \frac{(x-2)\cancel{(x+1)}}{\cancel{x+1}} = -3$$

(b) $$\lim_{x \to 1} \frac{(\sqrt{x+3} - 2)(\sqrt{x+3} + 2)}{(x-1)(\sqrt{x+3} + 2)} = \lim_{x \to 1} \frac{\overbrace{(x+3) - 4}^{x-1}}{(x-1)(\sqrt{x+3} + 2)} = \lim_{x \to 1} \frac{1}{\sqrt{x+3} + 2} = \frac{1}{\underbrace{\sqrt{4}}_{2} + 2} = \frac{1}{4}$$

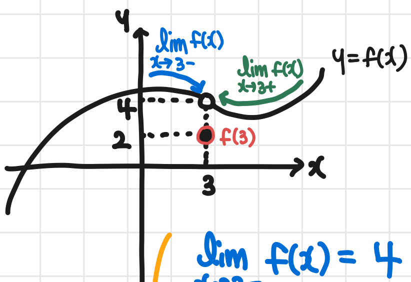
• 함수의 연속이 성립하려면,

① $f(a)$가 존재 (정의)

② $\lim\limits_{x \to a} f(x)$ 이 존재

③ $\lim\limits_{x \to a} f(x) = f(a)$

이면, 함수 $f(x)$가 $x=a$ 에서 연속.

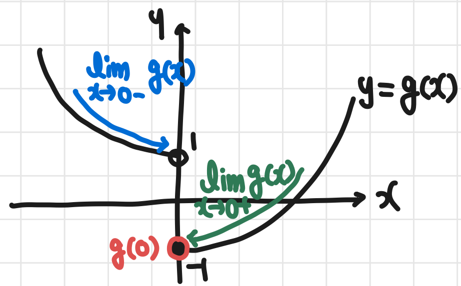



$\lim\limits_{x \to 3-} f(x) = 4$

$\lim\limits_{x \to 3+} f(x) = 4$

극한값有 ←

연속X ← $f(3) = 2$

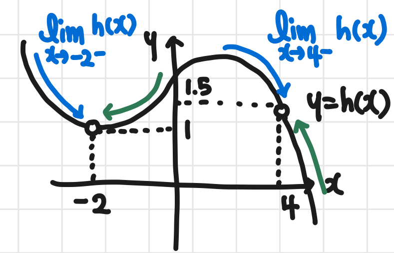



$\lim\limits_{x \to 0-} g(x) = 1$

$\lim\limits_{x \to 0+} g(x) = -1$

→ 극한값無

$g(0) = -1$ → 연속X (불연속)




$\lim\limits_{x \to -2-} h(x) = \lim\limits_{x \to -2+} h(x)$ $\cancel{h(-2)}$

연속X.

$\lim\limits_{x \to 4-} h(x) = \lim\limits_{x \to 4+} h(x)$ $h(4)$ X

연속X

- 구간
  - $a \underset{}{\overset{\text{초과}}{<}} x \overset{\text{미만}}{<} b \Leftrightarrow (a,b)$ ex) 구간 $(a,b)$ 에서 $f(x)$가 연속?
  - $a \le x < b \Leftrightarrow [a,b)$
  - $a < x \le b \Leftrightarrow (a,b]$
  - $a \overset{\text{이상}}{\le} x \overset{\text{이하}}{\le} b \Leftrightarrow [a,b]$

추가 실수 전체 : $(-\infty, \infty) = \mathbb{R}$

양수 전체 : $(0, \infty) = \mathbb{R}^+$

음수 전체 : $(-\infty, 0) = \mathbb{R}^-$

0이 포함된 양수 전체 : $[0, \infty) = \mathbb{R}^+$

" 음수 " : $(-\infty, 0]$

- 연속함수